SIMATS ENGINEERING

Lab Programs

COURSE CODE: CSA1418 **COURSE NAME: Compiler Design**

Experiment 10: C Program to Eliminate Left Factoring from a Given CFG

Aim:

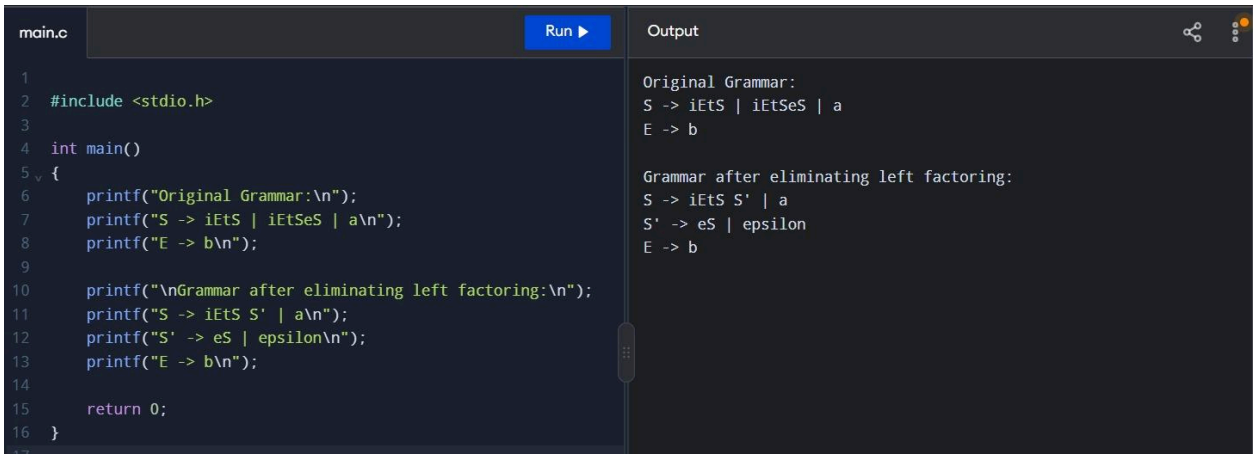
To write a C program to eliminate left factoring from the given context-free grammar (CFG).

Software Used: Programiz Online C Compiler.

Algorithm / Procedure:

- Open the Programiz Online C Compiler.
- Write the C program to display the grammar before and after eliminating left factoring.
- Consider the grammar:
  - S → iEtS | iEtSeS | a
  - E → b
- Identify the common prefix iEtS in the first two productions of S.
- Factor out the common prefix.
- Transform the productions into S → iEtSS′ | a and S′ → eS | ε.
- Display the original and transformed grammars.
- Compile and execute the program.
- Verify the output.

Execution Screenshot:



Result:

Thus, the C program was implemented and executed successfully to display the grammar after eliminating left factoring.